

Zhe "Eric" Zhang

zhe.eric.zhang@gmail.com • (804) 212-2486 • Richmond, VA • [LinkedIn](#)

Summary

Strategic fraud risk and data leader with hands-on expertise building end-to-end fraud detection and disposition solutions for one of the largest credit card issuers. Proven track record of combining advanced analytics, product development, and cross-functional execution to translate complex fraud challenges into scalable decision systems driving top-tier loss-to-sales performance at ~50% of the industry average while improving customer experience, operational efficiency, and profitability.

Professional Experience Highlights

Capital One Financial

Richmond, VA

Head of Data and Decision Support Services, Card Fraud

May 2006 – Present

— Progressed from data analyst to the leadership role

- Architected Fraud and Disputes data flows to streamline end-to-end workflows spanning authorization decisioning, concern logging, fraud treatment orchestration, customer notifications, and authentication eligibility determination over 30+ tools.
- Engineered batch and streaming feature pipelines for the Fraud Feature Factory, establishing a large-scale feature retrieval architecture 25 months ahead of the enterprise feature platform rollout; enabled real-time authorization scoring and decisioning over 35K TPS, with plenty of capacity to support Discover portfolio integration.
- Built enterprise customer data pipelines to power an in-house fraud link-analysis solution, clustering customers across shared attributes, enabling time-series fraud tagging, bulk remediation actions such as restrictions and card reissuance, and benchmarking against external network analysis vendors.
- Directed the data analytics team in a cross-functional partnership with US Card Fraud Intent, Operations, and Tech to drive cloud migration. Rationalized and decommissioned 90% of 300+ legacy reports and processes, while building cloud-native data solutions to empower analytics for 350+ business analysts and operational data needs for thousands of front-line agents.
- Designed and deployed foundational fraud testing to answer a core fraud decisioning question — when a decline creates net economic value — by quantifying tradeoffs among fraud loss prevention, customer friction, and downstream profitability; established the analytical basis for the Value-of-Delay framework used to value and calibrate card-present and card-not-present decline strategies.
- Delivered cross-functional data support beyond Fraud for Underwriting (Credit Line Decrease) and Recoveries asset sales; supported Recoveries Core Modernization through analytical data usage analysis and reconciliation against operational sources as Recoveries became the first area in Card migrated to the modern core platform.

Manager/Sr. Manager, Product Management

Apr 2018 – Jan 2020

— Dual role, in addition to managing the Fraud data analyst team

- Led product strategy and technical delivery for Big T, a modular data pipeline framework that reduced the handoff time of analyst-developed pipelines to Tech-managed production platforms from weeks to minutes. Developed reusable enterprise-grade modules for tokenization/de-tokenization, encryption/decryption, sensitive data detection/suppression, and secure file movement across multiple cloud environments.
- Launched Capital One's first cloud-based data producing pipeline, six months ahead of the enterprise Data Lake rollout, delivering near real-time data for a mission-critical customer service platform serving 100+ tenants. Directly enabled advanced analytics and operational staffing optimization across the platform.
- Built an authoritative data repository serving as the single source of truth for identifying true-positive fraud declines, directly powering transactional fraud model training, real-time decline and casing strategy, monitoring and valuation. Implemented point-in-time snapshots to prevent data leakage in model training and evaluation.

Education

Johns Hopkins University

M.S., Artificial Intelligence (In progress)

Baltimore, MD

Clemson University

M.S., Mechanical Engineering + Mathematical Sciences

Clemson, SC

Tsinghua University

B.S., Mechanical Engineering

Beijing, China

Core Skills

Fraud Strategy	Authorization Decisioning & Optimization, Casing Strategy, Authentication Strategy, Link Analysis
Data Strategy	Data Architecture, Product Management, Data Governance, Platform Modernization, Cloud Migration
ML/AI	Statistical Test Design, Feature Engineering, Classification, Anomaly Detection, Agentic AI Systems
Technical	Python, SQL, Spark, Git, Jira, AWS, Databricks, Dashboards, Data Visualization